Lets package it for a portfolio

Here's a **step-by-step guide to package your geospatial satellite analysis project for a professional portfolio**—so you impress recruiters, hiring managers, and anyone viewing your work:

Step 1: Write a Compelling Project Summary

Title:
Detecting Urban Expansion in Houston Using Landsat Satellite Imagery

Summary (80–120 words):
"This project analyzes urban area growth in Houston, TX over five years using open Landsat satellite imagery and geospatial indices. By calculating and visualizing the Normalized Difference Built-Up Index (NDBI), I automatically detect built-up (urban) pixels, map change hotspots, and quantify % expansion from 2018 to 2023. Outcomes provide actionable insights for city planners, environmental monitoring, and real estate stakeholders. Code, visuals, and result metrics are included for end-to-end reproducibility."

Step 2: Create and Organize Your GitHub Repository

Structure:

- `README.md` (with overview, goals, results, and links)
- `/notebooks` or `/src` (Jupyter notebook or script)
- `/images` or `/results` (maps, charts, before-after visuals)
- `requirements.txt` (list Python packages for reproducibility)

Sample README Outline:

```
# Detecting Urban Expansion in Houston via Satellite Imagery

## Summary
Analyze urban growth in Houston, TX from 2018 to 2023 using Landsat images and NDBI inde

## Method
- Download Landsat images for both years
- Calculate NDBI index to classify built-up area
- Compare, quantify, and visualize change

## Results
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- [Before & After Maps](/images/houston_growth.png)
- Urban area expanded by X% (2018–2023)
- [Chart: Urban Pixel Count by Year](/images/urban_pixel_chart.png)

## Technologies
Python, Google Earth Engine API, Jupyter Notebook, matplotlib/seaborn

## How to Reproduce
1. Download Landsat data (see `/data`)
2. Run `notebooks/urban_expansion_analysis.ipynb`
3. Review visuals in `/results`

## Impact
Supports smart city planning, environmental monitoring, and investment analysis.
```

Step 3: Visuals and Results

- Save 2–3 standout images (maps, charts) in `/images`.
- Include them in your README and possibly LinkedIn/portfolio posts.

Step 4: Write a LinkedIn/Portfolio Blurb

Example:
"Used Landsat satellite data and Python to detect and quantify urban expansion in Houston from 2018–2023. Delivered clear maps and charts showing % growth and new development hotspots—demonstrating real skills in geospatial reasoning, remote sensing, and applied data science."

Step 5: Share and Link Everywhere

- **GitHub:** Make repo public; pin to your "repositories" or highlight on your GitHub profile.
- **LinkedIn:**
  - Add to "Featured" section with a direct link and visuals
  - Mention in your Experience or About section
- **Portfolio Site:**
  - Create a "Projects" page with summary, images, and a "View Code" button to GitHub.

Step 6: (Optional) Demo Video

- Record a 1-minute walk-through of the notebook or final results, upload to YouTube, and link in your README/LinkedIn.

Ready to go? If you want help writing the README, designing the visuals, or a script for your demo video—just say so! I'll tailor those next.